Antisymmetric Floquet-Isospectral Potentials: Existence, Local Multiplicity, and Singular Period Inflations

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Abstract

For the discrete periodic Schrödinger operator on the integer lattice, we study the finite scheme of complex potentials that are antisymmetric under reversal and Floquet isospectral to zero. If the period is q and $m = \lfloor q/2 \rfloor$, we identify the associated graded local algebra at zero with $\mathbb{C}[x_1, \ldots, x_m]/(x_1^2, \ldots, x_m^2)$. Its length is therefore 2^m. Comparing this local length with the known global length $2^m m!$ proves the existence of a nonzero antisymmetric potential for every $q > 3$, answering Conjecture 1 of Faust, Friedman, O'Malley, Ramos, and Sharma. We also prove an exact differential-rank identity for period repetition. It produces a nonzero singular point at period 15 and inequivalent nonzero singular points at period 20, disproving the odd regularity and even uniqueness assertions of their Conjecture 3. The same argument gives arbitrarily many singular symmetry classes. The five theorem labels organize these results; they represent resolutions of two previously numbered conjectures, not five independent open problems. Historical priority is discussed separately from the proofs.

1 Targets, contribution, and the auxiliary questions

The prior targets are Conjectures 1 and 3 of [1], a preprint submitted on 9 February 2026. In the notation introduced below, their assertions are:

(i) For every integer $q > 3$, the antisymmetric zero-isospectral scheme $\mathcal{X}_q$ has a nonzero point (Conjecture 1).

(ii) At odd q, all nonzero points are regular; at even $q > 6$, there is one nonzero singular symmetry class, with multiplicity $2^{q/2-2}$; the origin has multiplicity $2^{\lfloor q/2 \rfloor}$ (Conjecture 3).

Here multiplicity and regularity refer to the defining spectral ideal, with its possible nilpotents. Also, a $q\mathbb{Z}$-periodic potential need not have least period q. These conventions matter for the counterexamples.

We introduce two auxiliary questions as the entry points to the argument. They are research questions formulated here, without an assertion that the questions themselves have never previously been considered.

Auxiliary problem 1 (The quadratic local-algebra question)**.** Can the lowest nonzero terms of the antisymmetric spectral equations be simultaneously identified with independent squares? If so, does this determine the entire associated graded local algebra at the zero potential?

Auxiliary problem 2 (The repeated-point differential question)**.** If a potential of period r is repeated to period $q = kr$, what happens to the rank of the differential of the antisymmetric spectral map?

Theorem T71 answers Problem 1; Theorem T72 applies it to the unchanged Conjecture 1. Theorem T73 answers Problem 2; Theorems T74–T75 apply it to the unchanged Conjecture 3. The method connects a local calculation to a global existence statement: cyclic correlations become Fourier squares, the squares determine local length, and the difference between global and local length forces nonzero solutions.

The transfer identities, Fourier transform, intersection theory, and regular sequence arguments used here are classical. In particular, repetition preserves Floquet isospectrality in prior work [2, Section 4.2, equation (12)]. Our proposed contribution is the exact local-algebra identification and the resulting conjecture resolutions, together with the differential-rank calculation. We do not assert that these basic tools are a historically new general principle. Section 8 records the scope of the priority check.

2 The precise spectral scheme

Fix $q \geq 3$ and $m = \lfloor q/2 \rfloor$. Throughout, indices of a q-tuple run from 0 to $q-1$, with cyclic indices when indicated. Define

$$\mathcal{A}_q = \{v \in \mathbb{C}^q : v_j = -v_{q-1-j}\}.$$

It has coordinates $u = (u_1, \dots, u_m)$ through

$$v = \begin{cases} (u_1, \dots, u_m, -u_m, \dots, -u_1), & q = 2m, \\ (u_1, \dots, u_m, 0, -u_m, \dots, -u_1), & q = 2m+1. \end{cases}$$

For $z \neq 0$, let $L_v(z)$ be the cyclic tridiagonal matrix with diagonal v, ones immediately above and below the diagonal, and corner entries $(L_v(z))_{0,q-1} = z^{-1}$ and $(L_v(z))_{q-1,0} = z$. The potentials considered here are complex; no nonzero real potential is being asserted.

Set

$$A_a(\lambda) = \begin{pmatrix} \lambda - a & -1 \\ 1 & 0 \end{pmatrix}, \qquad M_q(v;\lambda) = A_{v_{q-1}}(\lambda) \cdots A_{v_0}(\lambda), \qquad D_q(v;\lambda) = \operatorname{tr} M_q(v;\lambda). \quad (1)$$

The recurrence $f_{j+1} = (\lambda - v_j) f_j - f_{j-1}$ shows that the Floquet condition is the condition that z be an eigenvalue of M_q. Since $\det M_q = 1$, the usual monic characteristic-polynomial identity is

$$\det(\lambda I - L_v(z)) = D_q(v;\lambda) - z - z^{-1}. \quad (2)$$

It can also be checked directly by determinant expansion. Consequently v is Floquet isospectral to zero precisely when

$$D_q(v;\lambda) = D_q(0;\lambda) = 2T_q(\lambda/2). \quad (3)$$

Here T_q is the Chebyshev polynomial of the first kind. The last equality follows from Cayley–Hamilton for A_0, whose trace is λ and determinant is one.

The discriminant is invariant under cyclic permutation and reversal of the potential. Cyclic invariance follows from the trace; reversal follows from (2) and reversal of the Floquet matrix. In addition, conjugating each transfer matrix by $\operatorname{diag}(1, -1)$ gives

$$D_q(-v;-\lambda) = (-1)^q D_q(v;\lambda).$$

On $\mathcal{A}_q$, reversal equals negation. Thus $D_q(v;\lambda)$ has parity q in λ, and each coefficient is even as a polynomial in u. The leading monic term is independent of u. Therefore

$$F_q(u;\lambda) := D_q(v;\lambda) - D_q(0;\lambda) = \sum_{\ell=1}^{m} f_{q,\ell}(u) \lambda^{q-2\ell}. \quad (4)$$

Every $f_{q,\ell}$ has zero constant and linear terms. We use the schemes

$$I_q = (f_{q,1}, \dots, f_{q,m}) \subset \mathbb{C}[u_1, \dots, u_m], \qquad \mathcal{X}_q = \operatorname{Spec}(\mathbb{C}[u_1, \dots, u_m]/I_q). \tag{5}$$

These generators differ only by signs from the specialized spectral generators in [1], so this is the same ideal and the same scheme. Write $\mu_q(v) = \dim_{\mathbb{C}} \mathcal{O}_{\mathcal{X}_q,v}$ for its local length.

Proposition 2.1 (Global length; also [1], Theorem 4.6)**.** *The scheme $\mathcal{X}_q$ is a finite complete intersection of total length*

$$\operatorname{length}(\mathcal{X}_q) = 2^m m!.$$

Proof. The degree of $f_{q,\ell}$ in u is 2ℓ, and its highest homogeneous part is $(-1)^\ell e_\ell(u_1^2, \dots, u_m^2)$. Indeed, the terms of highest potential degree in the relevant characteristic coefficient come from the diagonal product

$$\prod_{j=0}^{q-1} (\lambda - v_j) = \lambda^{q-2m} \prod_{j=1}^{m} (\lambda^2 - u_j^2).$$

If all $e_\ell(u_1^2, \dots, u_m^2)$ vanish, then $\prod_j (t - u_j^2) = t^m$, so all $u_j = 0$. The homogenizations of the m equations therefore have no common projective zero on the hyperplane at infinity. Their intersection is zero dimensional: a positive dimensional projective component would meet that hyperplane. Since there are m hypersurfaces in $\mathbb{P}^m$ and their intersection is zero dimensional, they form a complete intersection. Projective Bézout gives length $\prod_{\ell=1}^{m} (2\ell) = 2^m m!$. All its points lie in the affine chart, proving the assertion. □

3 T71: the local algebra at the zero potential

Let $\zeta = e^{2\pi i/q}$ and introduce unnormalized Fourier coordinates

$$h_k(v) = \sum_{j=0}^{q-1} v_j \zeta^{-kj}. \tag{6}$$

On $\mathcal{A}_q$, direct reversal of the sum gives

$$h_0 = 0, \qquad h_{-k} = -\zeta^{-k} h_k. \tag{7}$$

Consequently $h_1, \dots, h_m$ form a linear coordinate system on $\mathcal{A}_q$: they determine all Fourier coefficients, and Fourier inversion determines v.

Theorem T71 (The quadratic local algebra)**.** *For every $q \geq 3$, the homogeneous quadratic parts of the m equations $f_{q,\ell}$ are an invertible linear combination of $h_1^2, \dots, h_m^2$. If $\mathfrak{m}$ is the maximal ideal at the origin, then*

$$\operatorname{gr}_{\mathfrak{m}} \mathcal{O}_{\mathcal{X}_q,0} \simeq \mathbb{C}[h_1, \dots, h_m]/(h_1^2, \dots, h_m^2). \tag{8}$$

Its Hilbert series is $(1+t)^m$, and

$$\mu_q(0) = 2^m. \tag{9}$$

We first compute the quadratic terms explicitly. Write $U_s = U_s(\lambda/2)$, where $U_0 = 1$, $U_1 = \lambda$, and $U_{s+1} = \lambda U_s - U_{s-1}$. Define cyclic correlations

$$C_d(v) = \sum_{j=0}^{q-1} v_j v_{j+d}, \qquad K_d(\lambda) = U_{d-1} U_{q-d-1} \quad (1 \leq d \leq m).$$

Lemma 3.1 (Quadratic discriminant)**.** *If $F_q^{[2]}$ denotes the homogeneous degree-two part in u, then*

$$F_q^{[2]}(v;\lambda) = \sum_{1\le d<q/2} C_d(v)K_d(\lambda) + \begin{cases} \frac{1}{2}C_m(v)K_m(\lambda), & q=2m, \\ 0, & q=2m+1. \end{cases} \tag{10}$$

The $K_1,\dots,K_m$ are linearly independent.

Proof. In the transfer product, replace the factors at positions $i<j$ by their linear variations $-v_iE_{11}$ and $-v_jE_{11}$. With $A=A_0$ and $d=j-i$, the resulting trace coefficient is

$$v_iv_j\,\mathrm{tr}\big(A^{q-d-1}E_{11}A^{d-1}E_{11}\big) = v_iv_jU_{q-d-1}U_{d-1},$$

because $(A^s)_{11}=U_s$. Summing over unordered pairs gives (10). Antipodal pairs in even period are counted twice by C_m, which explains the factor $1/2$.

The elementary Chebyshev product identity yields

$$K_1 = U_{q-2}, \qquad K_d - K_{d-1} = U_{q-2d} \quad (2\le d\le m). \tag{11}$$

For completeness, substitution $\lambda = 2\cos\theta$ and $U_s(\cos\theta)=\sin((s+1)\theta)/\sin\theta$ reduces the difference to the sine subtraction formula. Polynomial identity then extends it to all λ. The right sides of (11) have the distinct degrees $q-2, q-4,\dots,q-2m$, proving independence. □

Proof of the quadratic assertion in Theorem T71. Lemma 3.1 says that the quadratic coefficient forms span exactly the same space as $C_1,\dots,C_m$. We now identify that space. The correlations satisfy

$$C_{q-d}=C_d, \qquad \sum_{d=0}^{q-1} C_d = \left(\sum_j v_j\right)^2 = 0.$$

Thus C_0 is a linear combination of $C_1,\dots,C_m$. Taking the discrete Fourier transform in d gives

$$\sum_{d=0}^{q-1} C_d\zeta^{-kd} = h_kh_{-k} = -\zeta^{-k}h_k^2. \tag{12}$$

Every h_k^2, $1\le k\le m$, therefore belongs to the span of $C_1,\dots,C_m$. Conversely, Fourier inversion in (12), using (7), expresses each C_d in the span of $h_1^2,\dots,h_m^2$. Since these are squares of independent coordinates, they are linearly independent quadratic forms. Both spans have dimension m, so the change of generators is invertible. □

To pass from quadratic terms to the full local algebra, we record the precise filtered-algebra fact being used.

Lemma 3.2 (Regular lowest forms)**.** *Let $R=\mathbb{C}[[x_1,\dots,x_m]]$ and let $g_1,\dots,g_m\in R$ all have order d. Suppose their lowest homogeneous forms $Q_1,\dots,Q_m$ form a regular sequence. Then*

$$\mathrm{gr}(R/(g_1,\dots,g_m)) = \mathbb{C}[x_1,\dots,x_m]/(Q_1,\dots,Q_m).$$

Proof. The inclusion $(Q_1,\dots,Q_m)\subseteq \mathrm{in}(g_1,\dots,g_m)$ is immediate. For the reverse inclusion, take a nonzero $g=\sum_i a_ig_i$ and put $n=\min_i \mathrm{ord}(a_i)$. If $n+d<\mathrm{ord}(g)$, the degree-n parts A_i satisfy $\sum_i A_iQ_i=0$. First syzygies of a regular sequence are the Koszul syzygies, so there are homogeneous $B_{ij}=-B_{ji}$ of degree $n-d$ such that $A_i=\sum_j B_{ij}Q_j$. Replacing

$$a_i \quad\text{by}\quad a_i - \sum_j B_{ij}g_j$$

does not change $\sum_i a_i g_i$, since the B_{ij} are skew symmetric, and increases the minimum order of the coefficients. Repeat until $n + d = \operatorname{ord}(g)$. This requires only finitely many steps. The lowest part of g now belongs to $(Q_1, \dots, Q_m)$, as required. The fact about syzygies is the first exactness statement of the Koszul complex of a regular sequence. For an elementary induction, reduce a relation modulo the ideal of the first $m-1$ forms. The last form is a non-zero-divisor in that quotient, so its coefficient belongs to that ideal. Subtract the corresponding Koszul relations and induct on the remaining $m-1$ forms. □

Completion of the proof of Theorem T71. An invertible constant change of the equations makes their lowest forms $h_1^2, \dots, h_m^2$. These form a regular sequence: after quotienting by the preceding squares, multiplication by the next square remains injective on the free module over the remaining polynomial variables. Lemma 3.2 applies with $d = 2$. Completing a finite local $\mathbb{C}$-algebra does not change it, its associated graded algebra, or its length. This proves (8). The square-free monomials $\prod_i h_i^{\epsilon_i}$, $\epsilon_i \in \{0, 1\}$, form a basis of the graded quotient, giving Hilbert series $(1 + t)^m$ and length 2^m. □

Remark. The associated graded algebra is identified in (8); an isomorphism of the original, ungraded local algebra with that square algebra is not asserted. Moreover, existence in the next section needs only the weaker upper bound $\mu_q(0) \le 2^m$, which follows already from containment of the quadratic forms in the initial ideal.

4 T72: all-period antisymmetric existence

Theorem T72 (Conjecture 1 of [1])**.** *For every integer $q > 3$, there is a nonzero $v \in \mathcal{A}_q$ such that $L_v(z)$ and $L_0(z)$ have identical characteristic polynomials for every $z \neq 0$. More precisely, with $m = \lfloor q/2 \rfloor$,*

$$\sum_{v \in \mathcal{X}_q(\mathbb{C}),\, v \neq 0} \mu_q(v) = 2^m(m! - 1). \tag{13}$$

Proof. A finite algebra over $\mathbb{C}$ decomposes as the product of its local algebras. By Proposition 2.1, the sum of their lengths is $2^m m!$. Theorem T71 gives the contribution 2^m at zero. Subtracting proves (13). When $q > 3$, we have $m \ge 2$, so its right side is positive. Some nonzero point must therefore exist. By (4)–(5), all its discriminant coefficient differences vanish. Equation (2) gives the asserted Floquet isospectrality, with no restriction on the period q. □

This argument does not interpret the global length as a count of distinct points: it first removes the full local contribution of zero. That distinction is essential. For $q = 3$, $m = 1$ and the total length equals the origin length, so there is no nonzero antisymmetric point. Thus the threshold in the theorem is sharp for $q \ge 3$.

5 T73: exact rank under period repetition

For $w \in \mathcal{A}_r$ and $q = kr$, write $w^{[k]}$ for its concatenation k times. Reversal of a repeated word is the repetition of its reversal, so $w^{[k]} \in \mathcal{A}_q$. Define a linear map on variations by

$$S : \mathcal{A}_q \longrightarrow \mathcal{A}_r, \qquad (Sh)_j = \sum_{\ell=0}^{k-1} h_{j+\ell r} \quad (0 \le j < r). \tag{14}$$

Reversal of the indices shows $(Sh)_{r-1-j} = -(Sh)_j$. The map is onto, since $S(\eta^{[k]}) = k\eta$. Let $\Phi_q : \mathcal{A}_q \to \mathbb{C}^m$ be the coefficient map $(f_{q,1}, \dots, f_{q,m})$, and write $\rho_q(v) = \operatorname{rank}(d\Phi_q)_v$.

Theorem T73 (The repeated-point differential)**.** *For $r \geq 3$, $k \geq 1$, $q = kr$, and any $w \in \mathcal{A}_r$,*

$$D_q(w^{[k]}; \lambda) = 2T_k\big(D_r(w;\lambda)/2\big), \tag{15}$$

$$(dD_q)_{w^{[k]}}(h) = U_{k-1}\big(D_r(w;\lambda)/2\big)(dD_r)_w(Sh), \tag{16}$$

$$\rho_q(w^{[k]}) = \rho_r(w). \tag{17}$$

If $w \in \mathcal{X}_r(\mathbb{C})$, then $w^{[k]} \in \mathcal{X}_q(\mathbb{C})$ and

$$\dim_{\mathbb{C}} T_{w^{[k]}}\mathcal{X}_q = \lfloor q/2 \rfloor - \rho_r(w). \tag{18}$$

For $k \geq 2$, every such repeated point is singular.

Proof. The monodromy of a repeated potential is $M_r(w)^k$. For a determinant-one 2×2 matrix M, Cayley–Hamilton implies $\operatorname{tr} M^k = 2T_k(\operatorname{tr} M/2)$, proving (15).

Let τ be cyclic shift by r on $\mathbb{C}^q$ and let $P = k^{-1}\sum_{a=0}^{k-1}\tau^a$. The point $w^{[k]}$ is fixed by τ, and the discriminant is invariant under τ. Differentiating this invariance gives

$$(dD_q)_{w^{[k]}}(h) = (dD_q)_{w^{[k]}}(Ph).$$

The reversal operator R satisfies $R\tau R = \tau^{-1}$, so P commutes with R and preserves its negative eigenspace $\mathcal{A}_q$. Explicitly, $Ph = (Sh/k)^{[k]}$. Differentiating (15) along a repeated variation $\eta^{[k]}$, and using $T_k' = kU_{k-1}$, gives

$$(dD_q)_{w^{[k]}}(\eta^{[k]}) = kU_{k-1}(D_r(w)/2)(dD_r)_w(\eta).$$

Substitution of $\eta = Sh/k$ proves (16).

The factor $U_{k-1}(D_r(w;\lambda)/2)$ is a nonzero polynomial: D_r is monic of degree r, and U_{k-1} has degree $k-1$. Multiplication by it is injective in $\mathbb{C}[\lambda]$. Since S is onto, the images of the differentials have equal dimension. Reading coefficients in the appropriate parity polynomial spaces preserves that dimension, proving (17).

If w is zero-isospectral, the monodromy trace formula, or $T_k(T_r(x)) = T_{kr}(x)$, proves that its repetition is zero-isospectral. The Zariski tangent space of the scheme (5) is the kernel of $d\Phi_q$, yielding (18). When $k \geq 2$,

$$\rho_r(w) \leq \lfloor r/2 \rfloor < \lfloor kr/2 \rfloor.$$

The tangent space is nonzero. A zero-dimensional local ring over $\mathbb{C}$ is regular exactly when its tangent space is zero, so the point is singular. □

The trace identity (15) is not claimed as new. The precise content needed for the later applications is that the differential rank does not increase when the period is enlarged, although the antisymmetric ambient dimension does increase.

6 T74: an exact odd-period counterexample

We first give explicit regular seeds of periods four and five. Direct transfer multiplication gives, for arbitrary $a, b \in \mathbb{C}$,

$$D_4(a,b,-b,-a) - D_4(0) = -(a^2+b^2)\lambda^2 + a^2b^2 + a^2 - 2ab + b^2, \tag{19}$$

$$D_5(a,b,0,-b,-a) - D_5(0) = -(a^2+b^2)\lambda^3 + (a^2b^2 + 2a^2 - 2ab + b^2)\lambda. \tag{20}$$

Here and below the suppressed spectral argument is λ. Choose

$$Z = (1+i, 1-i, -1+i, -1-i), \qquad W = (a, ia, 0, -ia, -a), \quad a^2 = 1-2i. \tag{21}$$

Both choices of the square root a work. Equation (19) vanishes at Z: its quadratic coefficient is zero and its constant coefficient is $4-4=0$. For W, $b = ia$ gives $a^2+b^2=0$, and the remaining coefficient is

$$-a^4 + (1-2i)a^2 = 0.$$

Thus $Z \in \mathcal{X}_4$ and $W \in \mathcal{X}_5$. Both seeds, up to cyclic shift and complex conjugation, already occur in Papanicolaou [5, Example 3(ii)–(iii), pp. 807–809]; the period-four solution also appears in [3]. Neither seed is claimed as a new discovery.

Taking the two nontrivial coefficients in the order displayed, their Jacobian determinants with respect to (a, b) are

$$\det(d\Phi_4)_Z = -16i, \qquad \det(d\Phi_5)_W = -16+12i. \tag{22}$$

For example the first row is $(-2a, -2b)$; the second row for period five is $(2ab^2+4a-2b,\ 2a^2b-2a+2b)$. Substitution gives the second value in (22). Both seeds therefore have rank two. The least period of Z is four, since neither period one nor period two is possible. The least period of the nonconstant word W is five, since five is prime.

Theorem T74 (Odd singular points; refutation of Conjecture 3)**.** *With W as in (21), for every odd $k \geq 3$ the point $W^{[k]} \in \mathcal{X}_{5k}$ is nonzero and singular. Its differential rank is two and its Zariski tangent dimension is $(5k-1)/2-2$. In particular, $\mathcal{X}_{15}$ has a nonzero singular point with tangent dimension five. Hence the odd-period regularity assertion of Conjecture 3 in [1] is false.*

Proof. The seed identities and Theorem T73 prove every assertion. Explicitly, at period fifteen the word is

$$(a, ia, 0, -ia, -a,\ a, ia, 0, -ia, -a,\ a, ia, 0, -ia, -a).$$

Its seven antisymmetric coordinates are $(a, ia, 0, -ia, -a, a, ia)$, so it lies in the prescribed specialization exactly. Its discriminant is

$$D_{15} = D_5^3 - 3D_5, \qquad D_5 = \lambda^5 - 5\lambda^3 + 5\lambda,$$

which is the free discriminant. Its rank is $\rho_5(W) = 2 < 7$. The original statement quantifies over all $15\mathbb{Z}$-periodic potentials in this specialization, including those of smaller least period. Thus this example satisfies its hypotheses without changing the problem. □

7 T75: even uniqueness fails, and singular classes are unbounded

Use equivalence under cyclic shifts, reversal, global negation, and complex conjugation, including all their compositions. This is at least as broad as the symmetries used for the specialized scheme in [1]. Each of these operations preserves least period.

Theorem T75 (Multiple and unbounded singular classes)**.** *The scheme $\mathcal{X}_{20}$ contains the nonzero singular points*

$$P = Z^{[5]}, \qquad Q = W^{[4]},$$

where Z, W are defined in (21). Both have differential rank two and tangent dimension eight. They belong to different symmetry classes. Consequently, the even-period uniqueness assertion of Conjecture 3 in [1] is false.

More generally, for every integer $s \geq 1$ there is an even period q for which $\mathcal{X}_q$ has at least $s+1$ inequivalent nonzero singular points. For every $s \geq 2$ there is an odd period with at least s such classes.

Proof. Theorem T73 and (22) give the membership, rank, and tangent dimensions of P, Q. Their least periods are respectively four and five. Since every permitted symmetry preserves least period, they cannot be equivalent. This already disproves uniqueness at the allowed even period $20 > 6$.

For the general assertion, choose distinct primes $p_1, \ldots, p_s \geq 5$. Theorem T72 supplies a nonzero $w_j \in \mathcal{X}_{p_j}$ for each j. Such a word has least period p_j: the only smaller possible least period divides the prime p_j and hence is one, but a constant antisymmetric word must be zero. Let $q = 4p_1 \cdots p_s$. Repeat Z and each w_j to period q. Their repetition factors are all greater than one, so all these points are singular by Theorem T73. Their distinct least periods $4, p_1, \ldots, p_s$ make their symmetry classes distinct. For the odd assertion, take $q = p_1 \cdots p_s$ with $s \geq 2$ and repeat each w_j to period q. The same argument applies. □

Theorems T74 and T75 refute the two nonzero-point predictions in Conjecture 3. Theorem T71 proves its remaining origin-multiplicity assertion. No formula for the full local lengths of the nonzero singular points is asserted here; tangent dimension alone does not determine those lengths.

8 Scope, prior work, and verification

The mathematical conclusions in this manuscript are unconditional over $\mathbb{C}$. They depend on the displayed algebraic arguments, not on numerical root finding or on an unproved conjecture. Independent internal derivations checked the origin calculation both by cyclic correlations and by the band-edge argument in Appendix A. Exact symbolic checks of the low-period seeds and repeated-point Jacobians provide additional tests; they do not replace the general proofs. This manuscript has not undergone external peer review or proof-assistant verification.

Result	Precise contribution
T71	Full associated graded algebra and origin multiplicity, all $q \geq 3$.
T72	All-period affirmative solution of the existing Conjecture 1.
T73	Exact differential-rank identity for repetition.
T74	Period-15 counterexample and an infinite odd-period family.
T75	Period-20 failure of uniqueness and unbounded singular classes.

The prior global length calculation is explicitly credited in Proposition 2.1. Papanicolaou's 2021 examples [5] supply both small-period seeds, and the 2024 construction [3] gives a broader even-period family. The unrestricted existence theorem in [2] does not impose the reversal-antisymmetric constraint treated here, and period inflation in that paper is a prior ingredient. The more recent Jacobi-pencil work [4] concerns reducibility and diagonal equality patterns; no identical local-algebra identification was found in the inspected text.

A literature check through 24 October 2026 examined these primary papers, their available arXiv versions, associated author publication lists, and searches for the exact conjecture

numbers and the terms antisymmetric Floquet isospectrality, spectral-ideal multiplicity, and repeated-potential singularities. No earlier identical resolution was located in that search. This is a bounded search result, not a proof that no such work exists anywhere. In particular, this paper does not certify a historically unprecedented foundational principle.

Against the requested success conditions: the targets are pre-existing numbered conjectures and are used unchanged; the cited source presents them as open; complete arguments are supplied here; the local computation is not merely a renaming of the cited global count. The stronger priority and historical-new-principle requirements remain uncertified. Therefore this draft must not be advertised as having established every requested historical condition. It also makes no claim to settle a transcendence problem, Conjecture 2 of [1], or five separate named open problems.

A An independent band-edge proof of the quadratic structure

This appendix gives a second proof of the structural assertion in T71. It is useful for checking both the Fourier phase and the absence of a degenerate quadratic direction. For $1 \le k \le m$, put

$$\theta_k = \frac{\pi k}{q}, \qquad z_k = (-1)^k, \qquad \lambda_k = 2\cos\theta_k, \qquad c_k = h_k/q.$$

The real symmetric matrix $L_0(z_k)$ has the eigenvalue λ_k with multiplicity exactly two. An orthonormal basis of that eigenspace is

$$\varphi_+(j) = q^{-1/2} e^{i\theta_k j}, \qquad \varphi_-(j) = q^{-1/2} e^{-i\theta_k j}.$$

They obey the Floquet condition because $e^{\pm i\theta_k q} = z_k$. They are distinct and orthogonal since $1 \le k < q$. No other free eigenvector has the same eigenvalue, because equality of the corresponding cosines forces one of these two angles modulo 2π.

The compression of $\operatorname{diag}(v)$ to this eigenspace is

$$\begin{pmatrix} c_0 & c_k \\ c_{-k} & c_0 \end{pmatrix} = \begin{pmatrix} 0 & c_k \\ c_{-k} & 0 \end{pmatrix}.$$

Decompose $\lambda_k I - L_v(z_k)$ into this eigenspace and its orthogonal complement. Its complement block at $v = 0$ is invertible. The Schur complement therefore shows that the homogeneous quadratic term of its determinant equals

$$\gamma_k(-c_k c_{-k}) = \gamma_k \zeta^{-k} c_k^2,$$

where γ_k is the nonzero determinant of that complement block. The off-diagonal coupling correction has order two, and replacing an entry in the two-by-two leading block by that correction affects its determinant only in degree at least three. This justifies taking exactly the displayed quadratic term.

By (2), the determinant being expanded equals $F_q(v; \lambda_k)$, since the free determinant vanishes at λ_k. Evaluation at the m values λ_k is an invertible linear map on the polynomial space in (4). For even q, this is a Vandermonde map in the distinct numbers λ_k^2, including zero. For odd q, it is such a Vandermonde map followed by multiplication by the nonzero λ_k. Thus the quadratic coefficient forms are an invertible combination of the m independent squares c_k^2. Applying Lemma 3.2 gives the same local algebra and multiplicity.

B Exact computational checks and reproduction

The accompanying verifier uses symbolic polynomial arithmetic, not floating point approximations. Its checks include the following.

1. The discriminant identities (19) and (20) and the nonzero seed Jacobian determinants in (22).
2. Vanishing of the discriminant differences at $W^{[3]}$, $Z^{[5]}$, and $W^{[4]}$, computed directly at periods 15 and 20.
3. Rank two of the full specialized coefficient Jacobians at those repeated points, using exact arithmetic in the seed fields.
4. For small periods, agreement of the quadratic transfer expansion with (10), and length 2^m of the quotient by the homogeneous quadratic equations.

For the period-five seed, exact reduction modulo $a^2-(1-2i)$ suffices; for period four one may substitute $1+i$ directly or reduce modulo a^2-2i with $b=-ia$. In the latter case both choices of a give the stated ranks. Each Jacobian column can be computed without expanding all potential variables at once. If $B_j = A_{v_j}(\lambda)$ and $E = E_{11}$, then

$$\frac{\partial D_q}{\partial v_j} = -\operatorname{tr}(B_{q-1}\cdots B_{j+1}EB_{j-1}\cdots B_0).$$

The column for antisymmetric coordinate u_{j+1} is the difference between the v_j and v_{q-1-j} derivative polynomials. Extracting the coefficients $\lambda^{q-2\ell}$ produces the exact $m\times m$ Jacobian. This is an independent computational check of the rank identity, rather than a rank test on only repeated variations.

To compile this standalone source, run `pdflatex` twice on the `.tex` file. No external figure or bibliography file is required. The Python verifier and its machine-readable result are supplied as an optional verification bundle. Its finite checks certify only the cases executed; the all-period claims are established in the proofs above.

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